

Giseok Song

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Cheoncheon-dong, Jangan-gu,

Suwon-si, Gyeonggi-do, South Korea

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Education

Sungkyunkwan University

Undergraduate Chemical Engineering

02 2021 – Present

Suwon, Republic of Korea

Research

Undergraduate researcher at MDM Lab

Seoul National University

01 2026 – 02 2026

Seoul, Republic of Korea

- Process Optimization of Biodegradable Crack-Based Flexible Electronic Sensors
- Development and Process Optimization of Laser-Induced Sensors

Undergraduate researcher at PDSE Lab

Sungkyunkwan University

06 2025 – 08 2025

Suwon, Republic of Korea

- Enhancing the Accuracy of a Python-Based AI Model for E. coli Growth Rate Prediction
- Upstream Processing and Scale-Up of E. coli and Animal Cell Cultures

Researcher for Smart Factory Capstone Design

Sungkyunkwan University

03 2026 – 06 2026

Suwon, Republic of Korea

- GPS Framework: GlobalCLIP-PatchCore-SAM for Zero-Shot Anomaly Detection
- Overview: Three-stage zero-shot industrial anomaly detection pipeline integrating vision-language models with classical anomaly scoring and prompt-guided segmentation. Requires no task-specific training data, enabling direct deployment on unseen product categories
- Stage 1 – GlobalCLIP: Dual CLIP backbones (ViT-B/32 + ViT-L/14@336) with K=6 autoencoder ensemble. Adaptive Region Selection (ARS) and Adaptive Anomaly Weighting (AAW) modules compute coarse anomaly scores across global and local features
- Stage 2 – PatchCore (WideResNet-50): Patch-level feature extraction and nearest-neighbor anomaly scoring using a coreset-reduced memory bank for efficient inference.
- Stage 3 – SAM (ViT-H): Segment Anything Model refines anomaly localization via prompt-guided segmentation, producing pixel-accurate defect masks
- Benchmark: Evaluated on MVTec AD dataset (15 categories, 5,354 images). Achieves competitive image-level AUROC and pixel-level PRO scores under the zero-shot setting

Experience

Finance Manager

Sungkyunkwan University, Chemical Engineering Student Council

03 2025 – Present

Suwon, Republic of Korea

Manager

Sungkyunkwan University, Acoustic Band

03 2025 – 06 2025

Suwon, Republic of Korea

Awards & Honors

Academic Top Excellence Scholarship (Top 3%) <i>Sungkyunkwan University</i>	2026
President Dean's List <i>Sungkyunkwan University</i>	2025
Academic Top Excellence Scholarship (Top 7%) <i>Sungkyunkwan University</i>	2025
Academic Excellence Scholarship (Top 5%) <i>Sungkyunkwan University</i>	2022

Certifications

ADsP (Advanced Data Analytics Semi-Professional)
Industrial Engineer Hazardous Materials
Driver License (Class 1)
OPIc(Oral Proficiency Interview - computer) IH
JLPT(Japanese - Language Proficiency Test) N3

Patent Software

GPTCLIP: A Zero-shot Anomaly Detection and Localization Model for Industrial Environments Using GlobalCLIP Patch Tokens

Publications

GPT: GlobalCLIP Patch Token Framework for Zero-shot Industrial Anomaly Detection and Localization